

# Indranil Chatterjee

Kolkata, West Bengal



Portfolio



Indranil Chatterjee



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## Professional Summary

Co-Founder, CTO, and Full-stack AI Integrator specializing in architecting scalable, end-to-end web applications and high-performance systems. Experienced in building web apps from scratch to production, with a strong focus on responsive UI development, backend integration, and interactive user interfaces using React, JavaScript, and cloud services. Proficient across modern frontend frameworks (React, Tailwind CSS, Three.js) and robust backend architectures, bridging the gap between sophisticated AI pipelines and premium, user-centric digital ecosystems to solve complex engineering challenges.

## Skills & Achievements

- **Languages:** JavaScript (ES6+), TypeScript, Python, HTML5, CSS3
- **Frontend & Web Graphics:** React.js, Vite, Tailwind CSS, Three.js, Framer Motion, GSAP
- **Backend & Cloud:** Node.js, Firebase (Auth, Firestore, Hosting), REST APIs
- **Developer Tools:** Git, GitHub, Vercel, Linux
- **Core Concepts:** Data Structures, Algorithms, Object-Oriented Programming, MySQL
- **Achievements:** Rank 1 - Season 2 of the National Cloud & AI Innovation Challenge (Organized by 3SVK, Hyderabad)

## Top Projects

### Supermynd AI | EdTech Learning Platform

*React.js, Tailwind CSS, Node.js, Firebase, REST APIs*

*[Co-Founder & CTO]*

- Built and deployed the platform web app, handling UI layout, component state, and authentication.
- Connected the frontend to backend endpoints to stream generated study notes, flashcards, and quizzes directly to the browser.
- Set up custom domain routing, DNS records, and automated deployments on Vercel and Firebase.

### Universal LLM Tokenizer | Interactive Visualizer

*JavaScript (ES6+), HTML5, CSS3, DOM API*

*[Solo Project - Full-Stack Engineer]*

- Created an in-browser tool that breaks down raw input text into subwords and displays token boundaries in real time.
- Implemented the core byte-pair encoding algorithm entirely in vanilla JavaScript so it runs without servers or API keys.
- Added step-by-step merge tracking and an automatic token counter to estimate model pricing.

### AirScript | Gestural Communication App

*React.js, JavaScript, HTML5 Canvas, Web Speech API*

*[Solo Project - AI & Full-Stack Developer]*

- Built a web tool that tracks hand movement through a webcam to let users write words in the air.
- Parsed drawn coordinates on an HTML5 canvas and sent strokes to a vision model to convert gestures into clean text.
- Hooked into the browser Web Speech API to read out detected sentences out loud.

### ZeroTrace OSINT | Information Gathering Tool

*Python, JSON, DNS/WHOIS Protocols, REST APIs*

*[Solo Project - Backend Developer]*

- Wrote a command-line script to look up DNS entries, WHOIS records, and server IP details for target domains.
- Used concurrent requests in Python to query public registries simultaneously and speed up scan times.
- Formatted terminal output cleanly into readable tables and exported summary reports to JSON.

## Education

### B.Tech in Computer Science and Engineering (2025-2029)

JIS College of Engineering, Kalyani

Pursuing

## Interests

- Web development and open-source software.
- Hackathons and collaborative coding events.
- UI design and interactive web graphics.